Low-Rank Tensor Modeling for Hyperspectral Unmixing Accounting for Spectral Variability

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***Abstract*— Traditional hyperspectral unmixing methods neglect the underlying variability of spectral signatures often observed in typical hyperspectral images (HI), propagating these mismodeling errors throughout the whole unmixing process. Attempts to model material spectra as members of sets or as random variables tend to lead to severely ill-posed unmixing problems. Although parametric models have been proposed to overcome this drawback by handling endmember (EM) variability through generalizations of the mixing model, the success of these techniques depends on employing appropriate regularization strategies. Moreover, the existing approaches fail to adequately explore the natural multidimensinal representation of HIs. Recently, tensor-based strategies considered low-rank decompositions of HIs as an alternative to impose low-dimensional structures on the solutions of standard and multitemporal unmixing problems. These strategies, however, present two main drawbacks: 1) they confine the solutions to low-rank tensors, which often cannot represent the complexity of real-world scenarios and 2) they lack guarantees that EMs and abundances will be correctly factorized in their respective tensors. In this article, we propose a more flexible approach, called unmixing with low-rank tensor regularization algorithm accounting for EM variability (ULTRA-V), that imposes low-rank structures through regularizations whose strictness is controlled by scalar parameters. Simulations attest the superior accuracy of the method when compared with state-of-the-art unmixing algorithms that account for spectral variability.**

***Index Terms*— Endmember (EM) variability, hyperspectral data, low rank, tensor decomposition, unmixing with low-rank tensor regularization algorithm (ULTRA), ULTRA accounting for EM variability (ULTRA-V).**

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I. Introduction

HYPERSPECTRAL imaging has attracted formidable interest from the scientific community in the past two decades, and hyperspectral images (HI) have been explored in a vast, and increasing, number of applications in different fields [1]. The limited spatial resolution of hyperspectral devices often mixes the spectral contributions of different pure materials, termed *endmembers* (EM), in the scene [2]. This phenomenon is more explicit in remote sense applications due to the distance between airborne and spaceborne sensors and the target scene. The mixing process must be well understood for the vital information relating to the pure materials and their distribution in the scene to be accurately unveiled. Hyperspectral unmixing (HU) aims at solving this problem by decomposing the HI into a collection of EMs and their fractional *abundances* [3].

Different mixing models have been employed to explain the interaction between light and the EMs [1]–[4]. The simplest and most widely used model is the linear mixing model (LMM) [2], which assumes that the observed reflectance vector (i.e., a pixel) can be modeled as a convex combination of the spectral signatures of each EM present in the scene. Convexity imposes positivity and sum-to-one constraints on the linear combination coefficients. Hence, they represent the fractional abundances with which the EMs contribute to the scene. Although the simplicity of the LMM leads to fast and reliable unmixing strategies in some situations, it turns out to be simplistic to explain the mixing process in many practical applications. Hence, several approaches have been proposed in the literature to account for nonlinear mixing effects [4]–[6] and EM variability [7]–[9] often present in practical scenes.

A myriad of factors can induce EM variability, including environmental, illumination, atmospheric, and temporal changes [7]. If not properly considered, such variability can result in significant estimation errors being propagated throughout the unmixing process [10]. Most of the methods proposed so far to deal with spectral variability can be classified into three major groups: EMs as sets, EMs as statistical distributions, and, more recently, methods that incorporate the variability in the mixing model, often using physically motivated concepts [8]. The method proposed in this article belongs to the third group. Parametric models have received considerable interest since, unlike the other two approaches, they require neither spectral libraries to be known *a priori* nor strong hypothesis about the EMs statistical distribution.

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Recently, the variations of the LMM to cope with the spectral variability were introduced in [10]–[12]. Unmixing using these models lead to ill-posed problems that were solved by using a combination of different regularizations terms and variable splitting optimization strategies. The model perturbed LMM (PLMM) in [10] augmented the EM matrix with an additive perturbation matrix that needs to be estimated jointly with the abundances. Although the additive perturbation can model arbitrary EM variations, it is not physically motivated, and the excessive amount of degrees of freedom makes the problem even harder to solve. The extended LMM (ELMM) proposed in [11] introduces one new multiplicative term for each EM and can efficiently model changes in the observed reflectance due to illumination effects [11]. This model has a clear physical motivation, but its modeling capability is limited. The generalized LMM (GLMM) proposed in [12] generalizes the ELMM to account for variability in all regions of the measured spectrum. The GLMM is physically motivated and capable of modeling arbitrary variability, resulting in improved accuracy at the expense of a small increase in the computational complexity, when compared to the ELMM. Other works attempt to capture these complex spectral variations indirectly by means of additive residual terms [13], [14]. Although avoiding the interactions between the abundance fractions and the EM signatures, these strategies usually do not estimate the EM spectra for each image pixel.

The above-mentioned methods resort to different strategies to regularize the ill-posed optimization problem leading to the estimation of abundances and EMs. The regularization is achieved by introducing into the unmixing problem additional information based on common knowledge about the low dimensionality of structures embedded in HIs.

Possible ways to recover lower dimensional structures from noisy and corrupted data include the imposition of low-rank matrix constraints on the estimation process [15] or the low-rank decomposition of the observed data [16], [17]. The facts that HIs are naturally represented and treated as tensors and that low-rank decompositions of higher order (>2) tensors tend to capture homogeneities within the tensor structure make such strategies even more attractive for HU. Low-rank tensor models have been successfully employed in various tasks involving HIs, such as recovery of missing pixels [18], anomaly detection [19], classification [20], compression [21], dimensionality reduction [22], and analysis of multiangle images [23]. More recently, Veganzones *et al.* [23] and Qian *et al.* [24] considered low-rank tensor decompositions applied to standard and multitemporal HUs, respectively.

In [24], the HI is treated as a 3-D tensor, and spatial regularity is enforced through a nonnegative tensor factorization (NTF) strategy that imposes a low-rank tensor structure. In [23], nonnegative canonical polyadic decomposition (CPD) was used to unmix multitemporal HIs represented as 3-D tensors built by stacking multiple temporal matricized HIs. Although a low-rank tensor representation may naturally describe the regularity of HIs and abundance maps, the forceful introduction of stringent rank constraints may prevent an adequate representation of fast varying structures that are important for accurate unmixing. Another limitation of the approaches proposed in [23] and [24] is the lack of guarantee that EMs and abundances will be correctly factorized into their respective tensors. In [25], we proposed a new low-rank HU method called *unmixing with low-rank tensor regularization algorithm* (ULTRA), which accounts for highly correlated EMs. The HU problem was formulated using tensors and a low-rank abundance tensor regularization term was introduced. Differently, from the strict tensor decomposition considered in [23] and [24], ULTRA allowed important flexibility to the rank of the estimated abundance tensor to adequately represent fine-scale structure and details that lie beyond a low-rank structure but without compromising the regularity of the solution.

In this article, we extend the strategy proposed in [25] to account for the important effect of EM variability as well as a novel method to estimate the sufficient rank of a tensor for accurately solving the HI unmixing problem. The main novel contributions of this article are given as follows.

1) We extend the strategy proposed in [25] by imposing a new low-rank regularization on the 4-D EM tensor, which contains one EM matrix for each pixel, to account for EM variability. The new cost function results in an iterative algorithm, named *ULTRA accounting for EM variability* (ULTRA-V). At each iteration, ULTRA-V updates the estimations of the abundance and EM tensors as well as their low-rank approximations.
2) We propose a novel nontrivial strategy to determine the smallest rank representation that contains most of the variation of multilinear singular values [26].

Simulation results using synthetic and real data illustrate the performance improvement obtained using ULTRA-V when compared to competing methods, as well as its competitive computational complexity for relatively small images.

This article is organized as follows. Section II briefly reviews an important background on LMMs and definitions and notation used for tensors. Section III presents the proposed solution and the strategy to estimate tensor ranks. Section IV presents the simulation results and comparisons. Finally, Section V presents the conclusions.

II. Background and Notation

A. ELMMs

The LMM [2] assumes that a given pixel $\boldsymbol{r}_n = [r_{n,1}, \ldots, r_{n,L}]^\top$, with L bands, is represented as

$$\boldsymbol{r}_n = \boldsymbol{M}\boldsymbol{\alpha}_n + \boldsymbol{e}_n, \quad \text{s.t. } \mathbf{1}^\top\boldsymbol{\alpha}_n = 1 \text{ and } \boldsymbol{\alpha}_n \geq \mathbf{0} \tag{1}$$

where $\boldsymbol{M} = [\boldsymbol{m}_1, \ldots, \boldsymbol{m}_R]$ is an $L \times R$ matrix whose columns are the R EMs $\boldsymbol{m}_i = [m_{i,1}, \ldots, m_{i,L}]^\top$, $\boldsymbol{\alpha}_n = [\alpha_{n,1}, \ldots, \alpha_{n,R}]^\top$ is the abundance vector, $\boldsymbol{e} \sim \mathcal{N}(0, \sigma_n^2 \boldsymbol{I}_L)$ is an additive white Gaussian noise (WGN), $\boldsymbol{I}_L$ is the $L \times L$ identity matrix, and $\geq$ is the entrywise $\geqslant$ operator. The LMM assumes that the pure material EMs are fixed for all pixels $\boldsymbol{r}_n$, $n = 1, \ldots, N$, in the HI. This assumption can jeopardize the accuracy of estimated abundances in many circumstances due to the spectral variability existing in a typical scene.

Different extensions of the LMM have been recently proposed to mitigate this limitation. These models employ a

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different EM matrix for each pixel and are particular cases of the model

$$\boldsymbol{r}_n = \boldsymbol{M}_n \boldsymbol{\alpha}_n + \boldsymbol{e}_n \tag{2}$$

where $\boldsymbol{M}_n$ is the EM matrix for the nth pixel. Different parametric models propose different forms for $\boldsymbol{M}_n$ to account for spectral variability. These include additive perturbations over a mean matrix in the PLMM [10], multiplicative factors applied individually to each EM in the ELMM [11], or to each band in the GLMM [12]. Moreover, spatial regularization of the multiplicative scaling factors in the ELMM and GLMM helps to further mitigate the ill-posedness of the problem. Note that, although some works proposed to handle complex spectral variations indirectly by means of additive residual terms [13], [14], these strategies usually do not estimate the EM spectra for each image pixel and, like the other models, also require carefully designed regularization strategies.

Other approaches pursue different ways to improve the conditioning of the inverse problem, employing, for instance, multiscale regularization on the abundance maps [27], or using additional information in the form of spectral libraries known *a priori* [28], [29] or extracted from the observed HI [30].

All these models, however, fail to exploit the high-dimensional structure of the problem, which naturally suggests the representation of the HI, abundance maps, and EM matrices for all pixels as higher order tensors. In this article, instead of introducing a rigid parametric model for the EMs, we employ a more general tensor model, using a well-devised low-rank constraint to introduce regularity to the estimated EM tensor.

B. Notation

An order-P tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{N_1 \times \cdots \times N_P}$ $(P > 2)$ is an $N_1 \times \cdots \times N_P$ array with elements indexed by $\boldsymbol{\mathcal{T}}_{n_1,n_2,\ldots,n_P}$. The P dimensions of a tensor are called *modes*. A mode-ℓ fiber of tensor $\boldsymbol{\mathcal{T}}$ is the 1-D subset of $\boldsymbol{\mathcal{T}}$ obtained by fixing all but the ℓth dimension and is indexed by $\boldsymbol{\mathcal{T}}_{n_1,\ldots,n_{\ell-1},:,n_{\ell+1},\ldots,n_P}$. A slab or slice of tensor $\boldsymbol{\mathcal{T}}$ is a 2-D subset of $\boldsymbol{\mathcal{T}}$ obtained by fixing all but two of its modes. An HI is often conceived as a 3-D data cube and can be naturally represented by an order-3 tensor $\boldsymbol{\mathcal{R}} \in \mathbb{R}^{N_1 \times N_2 \times L}$, containing $N_1 \times N_2$ pixels represented by the tensor fibers $\boldsymbol{\mathcal{R}}_{n_1,n_2,:} \in \mathbb{R}^L$. Analogously, the abundances can also be collected in an order-3 tensor $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{N_1 \times N_2 \times R}$. Thus, given a pixel $\boldsymbol{\mathcal{R}}_{n_1,n_2,:}$, the respective abundance vector $\boldsymbol{\alpha}_{n_1,n_2}$ is represented by the mode-3 fiber $\boldsymbol{\mathcal{A}}_{n_1,n_2,:}$. Similarly, the EM matrices for each pixel can be represented as an order-4 tensor $\boldsymbol{\mathcal{M}} \in \mathbb{R}^{N_1 \times N_2 \times L \times R}$, where $\boldsymbol{\mathcal{M}}_{n_1,n_2,:,:} = \boldsymbol{M}_{n_1,n_2}$. We now review some operations of multilinear algebra (the algebra of tensors) that will be used in the rest of the article (more details can be found in [31]).

C. Tensor Product Definitions

Definition 1 (Outer Product): The outer product between vectors $\boldsymbol{b}^{(1)} \in \mathbb{R}^{N_1}, \boldsymbol{b}^{(2)} \in \mathbb{R}^{N_2}, \ldots, \boldsymbol{b}^{(P)} \in \mathbb{R}^{N_P}$ is defined as the order-P tensor $\boldsymbol{\mathcal{T}} = \boldsymbol{b}^{(1)} \circ \boldsymbol{b}^{(2)} \circ \cdots \circ \boldsymbol{b}^{(P)} \in \mathbb{R}^{N_1 \times N_2 \times \cdots \times N_P}$, where $\boldsymbol{\mathcal{T}}_{n_1,n_2,\ldots,n_P} = b^{(1)}_{n_1} b^{(2)}_{n_2} \cdots b^{(P)}_{n_P}$ and $b^{(i)}_{n_i}$ is the n_ith position of $\boldsymbol{b}^{(i)}$. It generalizes the outer product between two vectors.

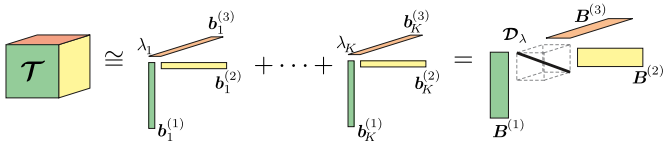


Fig. 1. Polyadic decomposition of a 3-D tensor, written as both outer products and mode-n products.

Definition 2 (Mode-k Product): The mode-k product, denoted $\boldsymbol{\mathcal{U}} = \boldsymbol{\mathcal{T}} \times_k \boldsymbol{B}$, of a tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{N_1 \times \cdots \times N_k \times \cdots \times N_P}$ and a matrix $\boldsymbol{B} \in \mathbb{R}^{M_k \times N_k}$ is evaluated such that each mode-k fiber of $\boldsymbol{\mathcal{T}}$ is multiplied by matrix $\boldsymbol{B}$, yielding $\boldsymbol{\mathcal{U}}_{n_1,n_2,\ldots,m_k,\ldots,n_P} = \sum_{i=1}^{N_k} \boldsymbol{\mathcal{T}}_{\ldots,n_{n-1},i,n_{n+1},\ldots} \boldsymbol{B}_{m_k,i}$.

Definition 3 (Multilinear Product): The full multilinear product, denoted by $[\![\boldsymbol{\mathcal{T}}; \boldsymbol{B}^{(1)}, \boldsymbol{B}^{(2)}, \ldots, \boldsymbol{B}^{(P)}]\!]$, consists of the successive application of mode-k products between $\boldsymbol{\mathcal{T}}$ and matrices $\boldsymbol{B}^{(i)}$, represented as $\boldsymbol{\mathcal{T}} \times_1 \boldsymbol{B}^{(1)} \times_2 \boldsymbol{B}^{(2)} \times_3 \ldots \times_P \boldsymbol{B}^{(P)}$.

Definition 4 (Mode-(M, 1) Contracted Product): The contracted mode-M product, denoted by $\boldsymbol{\mathcal{U}} = \boldsymbol{\mathcal{T}} \times^M \boldsymbol{b}$, is a product between a tensor $\boldsymbol{\mathcal{T}}$ and a vector $\boldsymbol{b}$ in mode-M, where the resulting singleton dimension is removed, given by $\boldsymbol{\mathcal{U}}_{\ldots,n_{n-1},n_{n+1},\ldots} = \sum_{i=1}^{N_n} \boldsymbol{\mathcal{T}}_{\ldots,n_{n-1},i,n_{n+1},\ldots} \boldsymbol{b}_i$.

D. CPD

An order-P rank-1 tensor is obtained as the outer product of P vectors. The rank of an order-P tensor $\boldsymbol{\mathcal{T}}$ is defined as the minimum number of order-P rank-1 tensors that must be added to obtain $\boldsymbol{\mathcal{T}}$ [17]. Thus, any tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{N_1 \times N_2 \times \cdots \times N_P}$ with $\operatorname{rank}(\boldsymbol{\mathcal{T}}) = K$ can be decomposed as a linear combination of at least K outer products of P rank-1 tensors. This so-called *polyadic decomposition* is illustrated in Fig. 1. When this decomposition involves exactly K terms, it is called the CPD [31] of a rank-K tensor $\boldsymbol{\mathcal{T}}$ and is given by

$$\boldsymbol{\mathcal{T}} = \sum_{i=1}^{K} \lambda_i \boldsymbol{b}_i^{(1)} \circ \boldsymbol{b}_i^{(2)} \circ \cdots \circ \boldsymbol{b}_i^{(P)}. \tag{3}$$

It has been shown that this decomposition is essentially unique under mild conditions [17]. The CPD can be written alternatively using mode-k products as

$$\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{D}}_\lambda \times_1 \boldsymbol{B}^{(1)} \times_2 \boldsymbol{B}^{(2)} \cdots \times_P \boldsymbol{B}^{(P)} \tag{4}$$

or using the full multilinear product as

$$\boldsymbol{\mathcal{T}} = [\![\boldsymbol{\mathcal{D}}_\lambda; \boldsymbol{B}^{(1)}, \boldsymbol{B}^{(2)}, \ldots, \boldsymbol{B}^{(P)}]\!] \tag{5}$$

where $\boldsymbol{\mathcal{D}}_\lambda = \operatorname{Diag}_P(\lambda_1, \ldots, \lambda_K)$ is the P-dimensional diagonal tensor and $\boldsymbol{B}^{(p)} = [\boldsymbol{b}_1^{(p)}, \ldots, \boldsymbol{b}_K^{(p)}]$, for $p = 1, \ldots, P$. Given a tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{N_1 \times N_2 \times \cdots \times N_P}$, the CPD can be obtained as the solution to the following optimization problem [17]:

$$\min_{\boldsymbol{\mathcal{D}}_\lambda, \boldsymbol{B}^{(1)}, \ldots, \boldsymbol{B}^{(K)}} \frac{1}{2} \left\| \boldsymbol{\mathcal{T}} - \sum_{i=1}^{K} \lambda_i \boldsymbol{b}_i^{(1)} \circ \cdots \circ \boldsymbol{b}_i^{(P)} \right\|_F^2. \tag{6}$$

A widely used strategy to compute an approximate solution to (6) is to use an alternating least-squares (LS) technique [17],

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which optimizes the cost function with respect to one term at a time, while keeping the others fixed, until convergence. Although optimization problem (6) is generally nonconvex, its solution is unique under relatively mild conditions, which is an important advantage of tensor-based methods [17].

E. Tensor Rank Bounds

Finding the rank of an arbitrary tensor $\mathcal{T}$ is NP-hard [32]. In [17], upper and lower bounds on tensor ranks are presented for arbitrary tensors. Let $\mathcal{T}$ be an order-3 tensor and

$$\begin{aligned} R_1 &\equiv \dim \operatorname{span}\{\mathcal{T}_{:,j,k}\}_{\forall j,k} \\ R_2 &\equiv \dim \operatorname{span}\{\mathcal{T}_{i,:,k}\}_{\forall i,k} \\ R_3 &\equiv \dim \operatorname{span}\{\mathcal{T}_{i,j,:}\}_{\forall i,j} \end{aligned} \tag{7}$$

be the mode-1 (column), mode-2 (row), and mode-3 (fiber) ranks, respectively, of $\mathcal{T}$. Thus, $K \equiv \operatorname{rank}(\mathcal{T})$ that is able to represent an arbitrary tensor is limited in the interval

$$\max(R_1, R_2, R_3) \leqslant K \leqslant \min(R_1 R_2, R_1 R_3, R_2 R_3). \tag{8}$$

The reader can note that the bounds presented above often lead to very large tensor ranks. In many practical applications, however, the "useful signal" rank is often much less than the actual tensor rank [17]. Hence, when low-rank decompositions are employed to extract low-dimensional structures from the signal, the ranks that lead to meaningful results are usually much smaller than $\max(R_1, R_2, R_3)$. In Section III-F, we propose a strategy to estimate the rank of tensor CPDs based on the variation of the multilinear singular values of $\mathcal{T}$.

III. Low-Rank Unmixing Problem

An effective strategy to capture the low-dimensional structures of HIs for solving the HU problem is to impose a low-rank structure to the abundance tensor [24]. The same strategy can also be applied to the EM tensor if one considers the EM variabilities to be small or highly correlated in low-dimensional structures within the HI. The low-rank property of HI tensors has been an important tool in the design of HI completion [33] and restoration algorithms [34], consisting in one of the main low-dimensional structures that are currently being considered in hyperspectral imaging applications. Thus, assuming that $\mathcal{A}$ has a low-rank $K_{\mathcal{A}}$, and that $\mathcal{M}$ has a low-rank $K_{\mathcal{M}}$, the global cost functional for the unmixing problem can be written as

$$\begin{aligned} J(\mathcal{A}, \mathcal{M}) = \frac{1}{2} \sum_{n_1=1}^{N_1} \sum_{n_2=1}^{N_2} \|\mathcal{R}_{n_1,n_2,:} - \mathcal{M}_{n_1,n_2,:,:} \mathcal{A}_{n_1,n_2,:}\|_F^2 \\ \text{s.t. } \operatorname{rank}(\mathcal{M}) = K_{\mathcal{M}}, \quad \mathcal{M} \geq \mathbf{0} \\ \operatorname{rank}(\mathcal{A}) = K_{\mathcal{A}}, \quad \mathcal{A} \geq \mathbf{0} \\ \mathcal{A} \times^3 \mathbf{1}_R = \mathbf{1}_{N_1 \times N_2}. \end{aligned} \tag{9}$$

Defining the HU problem as in (9) with fixed data-independent ranks $K_{\mathcal{M}}$ and $K_{\mathcal{A}}$ limits its flexibility to adequately represent the desired abundance maps and EM variability. Although fixing low ranks for $\mathcal{A}$ and $\mathcal{M}$ tends to capture the most significant part of the tensor energy [35], one may incur in a loss of fine and small-scale details that may be relevant for specific data. On the other hand, using large values for $K_{\mathcal{A}}$ and $K_{\mathcal{M}}$ makes the solution sensitive to noise, undermining the purpose of regularization. Thus, an important issue is how to effectively impose the low-rank constraint to achieve regularity in the solution without undermining its flexibility to adequately model small variations and details.

We propose to modify (9) by introducing new regularization terms, controlled by two low-rank tensors $\mathcal{Q} \in \mathbb{R}^{N_1 \times N_2 \times R}$ and $\mathcal{P} \in \mathbb{R}^{N_1 \times N_2 \times L \times R}$, to impose nonstrict constraints on $K_{\mathcal{A}}$ and $K_{\mathcal{M}}$. Doing that, tensors $\mathcal{Q}$ and $\mathcal{P}$ work as *a priori* information, and the strictness of the low-rank constraint is controlled by two additional parameters $\lambda_{\mathcal{A}}, \lambda_{\mathcal{M}} \in \mathbb{R}^+$. The proposed cost function is given by

$$\begin{aligned} J(\mathcal{A}, \mathcal{M}, \mathcal{P}, \mathcal{Q}) \\ = \frac{1}{2} \sum_{n_1=1}^{N_1} \sum_{n_2=1}^{N_2} \|\mathcal{R}_{n_1,n_2,:} - \mathcal{M}_{n_1,n_2,:,:} \mathcal{A}_{n_1,n_2,:}\|_F^2 \\ + \frac{\lambda_{\mathcal{M}}}{2} \|\mathcal{M} - \mathcal{P}\|_F^2 + \frac{\lambda_{\mathcal{A}}}{2} \|\mathcal{A} - \mathcal{Q}\|_F^2 \\ \text{s.t. } \mathcal{M} \geq \mathbf{0}, \quad \mathcal{A} \geq \mathbf{0}, \ \mathcal{A} \times^3 \mathbf{1}_R = \mathbf{1}_{N_1 \times N_2} \end{aligned} \tag{10}$$

with $\operatorname{rank}(\mathcal{P}) = K_{\mathcal{M}}$ and $\operatorname{rank}(\mathcal{Q}) = K_{\mathcal{A}}$. The optimization problem becomes

$$(\widehat{\mathcal{A}}, \widehat{\mathcal{M}}, \widehat{\mathcal{P}}, \widehat{\mathcal{Q}}) = \underset{\mathcal{A}, \mathcal{M}, \mathcal{P}, \mathcal{Q}}{\arg\min} \ J(\mathcal{A}, \mathcal{M}, \mathcal{P}, \mathcal{Q}). \tag{11}$$

To solve (11), we propose to find a local stationary point by minimizing (10) iteratively with respect to each variable. The resulting algorithm is termed the ULTRA-V and is presented in Algorithm 1. The intermediate steps are detailed in the following.

Algorithm 1 Global Algorithm for Solving (10)

Input : $\mathcal{R}$, $\lambda_{\mathcal{M}}$, $\lambda_{\mathcal{A}}$, $\mathcal{A}^{(0)}$, and $\mathcal{M}^{(0)}$.
Output: $\widehat{\mathcal{A}}$ and $\widehat{\mathcal{M}}$.

```
K_Q = estimateTensorRank(A^(0));
K_P = estimateTensorRank(M^(0));
Set i = 0 ;
while stopping criterion is not satisfied do
    i = i + 1 ;
    P^(i) = arg min_P  J(A^(i-1), M^(i-1), P) ;
    Q^(i) = arg min_P  J(A^(i-1), M^(i-1), Q) ;
    M^(i) = arg min_M  J(A^(i-1), M, P^(i), Q^(i)) ;
    A^(i) = arg min_A  J(A, M^(i), P^(i), Q^(i)) ;
end
return Â = A^(i), M̂ = M^(i);
```

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A. Solving With Respect to $\mathcal{A}$

To solve problem (11) with respect to $\mathcal{A}$, we use only the terms in (10) that depend on $\mathcal{A}$, leading to the cost function

$$\begin{aligned} J(\mathcal{A}) = & \frac{1}{2}\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2}\|\mathcal{R}_{n_1,n_2,:} - \mathcal{M}_{n_1,n_2,:,:}\mathcal{A}_{n_1,n_2,:}\|_F^2 \\ & + \frac{\lambda_{\mathcal{A}}}{2}\|\mathcal{A}-\mathcal{Q}\|_F^2 \\ & \text{s.t. } \mathcal{A}\geq \mathbf{0}, \quad \mathcal{A}\times^3 \mathbf{1}_R = \mathbf{1}_{N_1\times N_2} \end{aligned} \tag{12}$$

which results in a standard regularized fully constrained LS (FCLS) problem that can be solved efficiently.

B. Solving With Respect to $\mathcal{M}$

Analogously to Section III-A, to solve problem (11) with respect to $\mathcal{M}$, we use only the terms in (10), which depend on $\mathcal{M}$, leading to

$$\begin{aligned} J(\mathcal{M}) = & \frac{1}{2}\sum_{n_1=1}^{N_1}\sum_{n_2=1}^{N_2}\|\mathcal{R}_{n_1,n_2,:} - \mathcal{M}_{n_1,n_2,:,:}\mathcal{A}_{n_1,n_2,:}\|_F^2 \\ & + \frac{\lambda_{\mathcal{M}}}{2}\|\mathcal{M}-\mathcal{P}\|_F^2 \\ & \text{s.t. } \mathcal{M}\geq \mathbf{0} \end{aligned} \tag{13}$$

which results in a regularized nonnegative LS problem. An approximate solution can be obtained ignoring the positivity constraint over the EM tensor and projecting the LS result onto the positive orthant as [12]

$$\begin{aligned} \widehat{\mathcal{M}}_{n1,n2,:,:} = \mathbb{P}_+\big(&\big(\mathcal{R}_{n1,n2,:}\mathcal{A}_{n1,n2,:}^\top + \lambda_{\mathcal{M}}\mathcal{P}_{n1,n2,:,:}\big) \\ & \times \big(\mathcal{A}_{n1,n2,:}\mathcal{A}_{n1,n2,:}^\top + \lambda_{\mathcal{M}}\boldsymbol{I}\big)^{-1}\big) \end{aligned} \tag{14}$$

where $\mathbb{P}_+ : \mathbb{R}^{N_1\times N_1\times L} \to \mathbb{R}_+^{N_1\times N_1\times L}$ is the projection operator that maps every negative element to zero. Although this solution is approximate, it is significantly faster than directly solving (13) and the algorithm still demonstrated a good empirical convergence in our experiments.

C. Solving With Respect to $\mathcal{P}$

Rewriting the terms in (10) that depend on $\mathcal{P}$ leads to

$$J(\mathcal{P}) = \frac{\lambda_{\mathcal{M}}}{2}\|\mathcal{M}-\mathcal{P}\|_F^2. \tag{15}$$

Assuming that most of the energy of $\mathcal{M}$ lies in a low-rank structure, we write the tensor $\mathcal{P}$ as a sum of a small number $K_{\mathcal{P}}$ of rank-1 components, such that

$$\mathcal{P} = \sum_{i=1}^{K_{\mathcal{P}}}\delta_i \boldsymbol{x}_i^{(1)}\circ\boldsymbol{x}_i^{(2)}\circ\boldsymbol{x}_i^{(3)}\circ\boldsymbol{x}_i^{(4)}. \tag{16}$$

This introduces a low-rank *a priori* condition on $\mathcal{P}$ whose strictness can be controlled by the regularization constant $\lambda_{\mathcal{P}}$. Using (16) in (15) leads to the optimization problem

$$\begin{aligned} &(\widehat{\boldsymbol{\Delta}}, \widehat{\boldsymbol{X}}^{(1)}, \widehat{\boldsymbol{X}}^{(2)}, \widehat{\boldsymbol{X}}^{(3)}, \widehat{\boldsymbol{X}}^{(4)}) \\ &= \underset{\boldsymbol{\Delta},\boldsymbol{X}^{(1)},\boldsymbol{X}^{(2)},\boldsymbol{X}^{(3)},\boldsymbol{X}^{(4)}}{\arg\min} \frac{\lambda_{\mathcal{M}}}{2} \\ &\quad \times \left\|\mathcal{M} - \sum_{i=1}^{K_{\mathcal{P}}}\delta_i \boldsymbol{x}_i^{(1)}\circ\boldsymbol{x}_i^{(2)}\circ\boldsymbol{x}_i^{(3)}\circ\boldsymbol{x}_i^{(4)}\right\|_F^2 \end{aligned} \tag{17}$$

where $\boldsymbol{\Delta} = \mathrm{Diag}_4(\delta_1,\ldots,\delta_{K_{\mathcal{P}}})$ is a 4-D diagonal tensor with $\boldsymbol{\Delta}_{i,i,i,i} = \delta_i$. Problem (17) can be solved using an alternating LS strategy [17].

Finally, the solution $\widehat{\mathcal{P}}$ is obtained from $\widehat{\boldsymbol{\Delta}}$, $\widehat{\boldsymbol{X}}^{(1)}$, $\widehat{\boldsymbol{X}}^{(2)}$, $\widehat{\boldsymbol{X}}^{(3)}$, and $\widehat{\boldsymbol{X}}^{(4)}$ using the full multilinear product as

$$\widehat{\mathcal{P}} = [\![\widehat{\boldsymbol{\Delta}}; \widehat{\boldsymbol{X}}^{(1)}, \widehat{\boldsymbol{X}}^{(2)}, \widehat{\boldsymbol{X}}^{(3)}, \widehat{\boldsymbol{X}}^{(4)}]\!]. \tag{18}$$

D. Solving With Respect to $\mathcal{Q}$

Analogous to Section III-C, the cost function to be optimized for $\mathcal{Q}$ can be written as

$$J(\mathcal{Q}) = \frac{\lambda_{\mathcal{A}}}{2}\|\mathcal{A}-\mathcal{Q}\|_F^2. \tag{19}$$

Assuming that most of the energy of $\mathcal{A}$ lies in a low-rank structure, we write tensor $\mathcal{Q}$ as a sum of a small number $K_{\mathcal{Q}}$ of rank-1 components, such that

$$\mathcal{Q} = \sum_{i=1}^{K_{\mathcal{Q}}}\xi_i \boldsymbol{z}_i^{(1)}\circ\boldsymbol{z}_i^{(2)}\circ\boldsymbol{z}_i^{(3)}. \tag{20}$$

This introduces a low-rank *a priori* condition on $\mathcal{A}$, which will be more or less enforced depending on the regularization constant $\lambda_{\mathcal{A}}$. Using (20) in (19) leads to the optimization problem

$$\begin{aligned} &(\widehat{\boldsymbol{\Xi}}\ ,\ \widehat{\boldsymbol{Z}}^{(1)}, \widehat{\boldsymbol{Z}}^{(2)}, \widehat{\boldsymbol{Z}}^{(3)}) \\ &= \underset{\boldsymbol{\Xi},\boldsymbol{Z}^{(1)},\boldsymbol{Z}^{(2)},\boldsymbol{Z}^{(3)}}{\arg\min} \frac{\lambda_{\mathcal{A}}}{2}\left\|\mathcal{A} - \sum_{i=1}^{K_{\mathcal{Q}}}\xi_i \boldsymbol{z}_i^{(1)}\circ\boldsymbol{z}_i^{(2)}\circ\boldsymbol{z}_i^{(3)}\right\|_F^2 \end{aligned} \tag{21}$$

where $\boldsymbol{\Xi} = \mathrm{Diag}_3(\xi_1,\ldots,\xi_{K_{\mathcal{Q}}})$ is an order-3 diagonal tensor with $\boldsymbol{\Xi}_{i,i,i} = \xi_i$. Problem (21) can be solved using an alternating LS strategy [17]. Finally, the solution $\widehat{\mathcal{Q}}$ is obtained from $\widehat{\boldsymbol{\Xi}}$, $\widehat{\boldsymbol{Z}}^{(1)}$, $\widehat{\boldsymbol{Z}}^{(2)}$, and $\widehat{\boldsymbol{Z}}^{(3)}$ using the full multilinear product as

$$\widehat{\mathcal{Q}} = [\![\widehat{\boldsymbol{\Xi}}\ ;\ \widehat{\boldsymbol{Z}}^{(1)}; \widehat{\boldsymbol{Z}}^{(2)}; \widehat{\boldsymbol{Z}}^{(3)}]\!]. \tag{22}$$

E. Computational Complexity of Algorithm 1

The computational complexity of each iteration of Algorithm 1 can be measured as follows. The optimizations with respect to $\mathcal{A}$ and $\mathcal{M}$ both consist of regularized constrained LS problems with $N_1N_2\ R$ and N_1N_2LR variables respectively. Thus, these problems can be solved with a complexity of $\mathcal{O}((N_1N_2\ R)^3)$ and $\mathcal{O}((N_1N_2\ LR)^3)$, respectively. The optimizations with respect to variables $\mathcal{P}$ and $\mathcal{Q}$ consist of CPDs of these tensors with ranks $K_{\mathcal{P}}$ and $K_{\mathcal{Q}}$, respectively. Considering an alternating LS (ALS) approach for the CPD, these optimization problems will have computational complexities of $\mathcal{O}(K_{\text{iter}}K_{\mathcal{P}}N_1N_2LR)$ and $\mathcal{O}(K_{\text{iter}}K_{\mathcal{Q}}N_1N_2R)$, respectively, where K_{iter} is the number of ALS iterations [36]. Thus, the overall complexity of the algorithm scales linearly with the number of ALS iterations and with the tensor ranks, and cubically in the problem dimensions. When processing large data sets, the extra complexity could be partially mitigated by applying image segmentation or band Selection strategies [6], [37], [38]. This analysis is beyond the scope of this article and will be addressed in the future.

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F. Estimating Tensor Ranks

In Section II-E, we have recalled important results relating bounds for order-3 tensor ranks to the span of the matricized versions of tensors. We have also noted, from our own experience, that those bounds tend to indicate tensor ranks that are larger than the rank associated with the information relevant for HU. Our interest in HU is to model low-dimensional structures of the HI using low-rank tensors. At the same time, this low-rank representation should be rich enough to include all dimensions of the original HI tensor that contains relevant information. Therefore, although the literature presents many rank estimation strategies (see [39] and references therein), in this article, we exploit the rank bounds discussed in Section II-E to approximate the "useful rank" of a tensor by the number of the largest singular values of their matricized versions required to represent most of the tensor energy.

Let $\boldsymbol{T}_i = \text{mat}_i(\boldsymbol{\mathcal{T}}) \in \mathbb{R}^{N_i \times (N_1 \ldots N_{i-1} N_{i+1}, \ldots N_P)}$ be the matricization of an arbitrary tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{N_1 \times N_2 \ldots \times N_P}$ obtained by stacking all tensor fibers along the i-th tensor dimension. Let $\boldsymbol{s}_i = \text{SVD}(\boldsymbol{T}_i)$ be the set of singular values of $\boldsymbol{T}_i$, sorted descending in value. Also, let $\boldsymbol{d}_i = \text{diff}(\boldsymbol{s}_i)$ be the vector of first order differences of the elements of $\boldsymbol{s}_i$, such that, $d_j^{(i)} = s_j^{(i)} - s_{j+1}^{(i)}$. Then, we define the ith candidate for rank of $\boldsymbol{\mathcal{T}}$ as the smallest index j such that $|d_j^{(i)}|$ sufficiently small, namely,

$$\hat{R}_i = \min j, \quad \text{s.t.,} \quad \left|d_j^{(i)}\right| < \varepsilon \tag{23}$$

where ε is a parameter limiting the singular value variation. In all experiments reported here, we used $\varepsilon = 0.15$. We have experimentally verified that the resulting abundance mean-squared error (MSE) has very low sensitivity to the choice of ε. Finally, we approximate the rank of tensor $\boldsymbol{\mathcal{T}}$ as

$$K = \max\{\hat{R}_1, \ldots, \hat{R}_P\}. \tag{24}$$

For the experiments reported in this article, we have used definition (24) to estimate $K_{\boldsymbol{\mathcal{P}}}$ and $K_{\boldsymbol{\mathcal{Q}}}$ in (16) and (20) from the abundance and EM tensors estimated using simple unmixing strategies such as the scaled constrained LS (SCLS) [11].

IV. Simulations

In this section, the performance of the proposed methodology is illustrated through simulations with both synthetic and real data. We compare the proposed ULTRA-V method with the FCLS, SCLS [11], PLMM [10], ELMM [11], and GLMM [12]. To highlight the differences between ULTRA-V and ULTRA [25], we also consider ULTRA for simulations with synthetic data.

To measure the accuracy of the unmixing methods, we consider the MSE

$$\text{MSE}_{\boldsymbol{\mathcal{X}}} = \frac{1}{N_{\boldsymbol{\mathcal{X}}}} \|\text{vec}(\boldsymbol{\mathcal{X}}) - \text{vec}(\hat{\boldsymbol{\mathcal{X}}})\|^2 \tag{25}$$

where $\text{vec}(\cdot)$ is the vectorization operator $\boldsymbol{\mathcal{X}} \to \boldsymbol{x}$, $\mathbb{R}^{a \times b \times c} \mapsto \mathbb{R}^{abc}$, $N_{\boldsymbol{\mathcal{X}}} = abc$, and the spectral angle mapper for the HI

$$\text{SAM}_{\boldsymbol{\mathcal{R}}} = \frac{1}{N} \sum_{n=1}^{N} \arccos\left(\frac{\boldsymbol{r}_n^\top \hat{\boldsymbol{r}}_n}{\|\boldsymbol{r}_n\| \|\hat{\boldsymbol{r}}_n\|}\right) \tag{26}$$

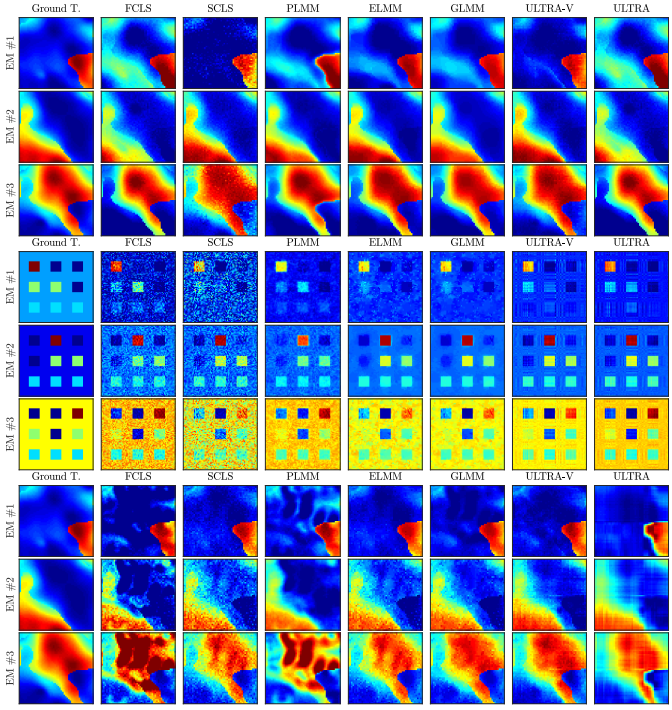


Fig. 2. Abundance maps of (top–down) DC0, DC1, and DC2 for all tested algorithms. Abundance values represented by colors ranging from blue ($\alpha_k = 0$) to red ($\alpha_k = 1$).

and for the EMs tensor

$$\text{SAM}_{\boldsymbol{\mathcal{M}}} = \frac{1}{N} \sum_{n=1}^{N} \sum_{k=1}^{R} \arccos\left(\frac{\boldsymbol{m}_{k,n}^\top \hat{\boldsymbol{m}}_{k,n}}{\|\boldsymbol{m}_{k,n}\| \|\hat{\boldsymbol{m}}_{k,n}\|}\right). \tag{27}$$

All the algorithms were implemented in MATLAB on a desktop computer equipped with an Intel Core I7 processor with 4.2 GHz and 16 GB of RAM. In all cases, we used EMs extracted using the vertex component analysis (VCA) [40] either to build the reference EM matrix or to initialize the different methods, with the number of EMs R assumed to be known *a priori*. The abundance maps were initialized using the maps estimated by the SCLS.

A. Synthetic Data

For a comprehensive comparison among the different methods, we created three synthetic data sets, namely, Data Cube 0 (DC0), Data Cube 1 (DC1), and Data Cube 2 (DC2), with 50×50 pixels (DC0 and DC2) and 70×70 pixels (DC1). DC0 and DC1 were built using three 224-band EMs extracted from the USGS Spectral Library [41], while DC2 was built using three 16-band minerals often found in bodies of the Solar System. For the three data sets, spatially correlated abundance maps were used, as depicted in the first column in Fig. 2. For DC0, we adopted the variability model used in [11] (a multiplicative factor acting on each EM). For DC1, we used the variability model according to the PLMM [10]. For DC2, we used the Hapke model [42] devised to realistically represent the spectral variability introduced due to changes in the illumination conditions caused by the topography of the scene [11]. WGN was added to all data sets to yield a 30-dB SNR.

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TABLE I
SIMULATION RESULTS USING SYNTHETIC DATA

	$\text{MSE}_{\mathcal{A}}$	$\text{MSE}_{\mathcal{M}}$	$\text{SAM}_{\mathcal{M}}$	$\text{MSE}_{\mathcal{R}}$	$\text{SAM}_{\mathcal{R}}$	Time
Data Cube 0 – DC0						
FCLS	1.81	-	-	16.11	5.92	0.42
SCLS	0.68	175.79	6.19	59.38	5.42	0.38
PLMM	0.76	**94.57**	**5.58**	4.89	3.41	81.89
ELMM	0.35	106.17	5.63	4.78	3.40	17.15
GLMM	**0.34**	101.51	5.87	**5.7e-3**	**0.09**	20.23
ULTRA-V	**0.23**	**92.39**	**5.56**	**0.73**	**1.32**	14.46
ULTRA	1.81	-	-	16.11	5.92	1.42
Data Cube 2 – DC1						
FCLS	2.01	-	-	6.93	3.71	0.74
SCLS	2.07	92.16	**5.37**	24.78	3.41	0.76
PLMM	1.58	157.00	8.49	2.75	2.41	120.48
ELMM	1.29	69.15	5.95	**0.01**	**0.09**	23.18
GLMM	1.20	**68.11**	6.09	**0.01**	**0.08**	29.81
ULTRA-V	**1.12**	**60.12**	**5.21**	4.26	2.56	29.47
ULTRA	**1.17**	-	-	10.93	3.94	4.44
Data Cube 3 – DC2						
FCLS	1.90	-	-	2.03	13.69	0.30
SCLS	0.71	**1.66**	**2.29**	**1.07**	12.68	0.30
PLMM	1.27	2.45	**2.28**	2.12	12.17	61.13
ELMM	0.63	2.84	3.32	2.31	13.13	11.15
GLMM	**0.59**	**1.84**	2.79	2.04	**12.58**	11.60
ULTRA-V	**0.46**	2.91	2.94	**9e-5**	**0.49**	5.19
ULTRA	0.83	-	-	1.02	3.73	0.36

To select the optimal parameters for each algorithm, we performed grid searches for each data set. We used parameter search ranges based on the ranges tested and discussed by the authors in the original publications. For the PLMM, we used $\gamma = 1$, since the authors fixed this parameter in all simulations and searched for α and β in the ranges $[10^{-6}, 10^{-3}, 0.1, 0.35, 0.7, 1.4, 5, 25]$ and $[10^{-9}, 10^{-5}, 10^{-4}, 10^{-3}]$, respectively. For both ELMM and GLMM, we selected the parameters among the following values: λ_S, $\lambda_M \in [0.01, 0.1, 1, 5, 10, 15]$, $\lambda_A \in [10^{-6}, 10^{-3}, 0.01, 0.05, 0.1, 1, 10,$ and λ_ψ, $\lambda_\Psi \in [10^{-6}, 10^{-3}, 10^{-1}]$, while for the proposed ULTRA-V, we selected the parameters in the intervals $\lambda_{\mathcal{A}} \in [0.001, 0.01, 0.1, 1, 10, 100]$ and $\lambda_{\mathcal{M}} \in [0.1, 0.2, 0.4, 0.6, 0.8, 1]$. For the ULTRA, we searched $\lambda_{\mathcal{A}}$ in the same interval used for the ULTRA-V.

The results are given in Table I, where the best and second best results for each metric are marked in bold red and bold blue, respectively. ULTRA-V clearly outperformed the competing algorithms for all data sets in terms of $\text{MSE}_{\mathcal{A}}$. For the other metrics, the best results depended on the data sets. In terms of $\text{MSE}_{\mathcal{M}}$ and $\text{SAM}_{\mathcal{M}}$, ULTRA-V yielded the best results for DC0 and DC1. Finally, ULTRA-V results for $\text{MSE}_{\mathcal{R}}$ and $\text{SAM}_{\mathcal{R}}$ were the second best for DC0 and the best for DC2. The execution times, shown in the rightmost columns in Table I, show that ULTRA-V required the smallest execution time among the more sophisticated algorithms (PLMM, ELMM, and GLMM) for DC0 and DC2 and comparable execution time for DC1. As expected, the ULTRA method

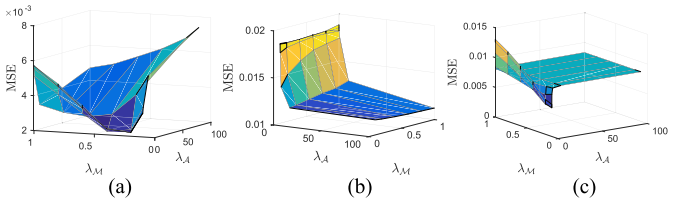


Fig. 3. Parameters sensitivity to changes around the optimal values. (a) Left: DC0. (b) Middle: DC1. (c) Right: DC2.

provided results that were often better than the FCLS but significantly worse than those obtained from methods accounting for EM variability. This happens because ULTRA imposes a low-rank structure over the abundances but keeps the EMs fixed for all pixels, what greatly limits the algorithm capacity to adapt to EM variations along the image.

1) Parameters Sensitivity: While we have proposed a strategy to determine rank values for tensors $\mathcal{P}$ and $\mathcal{Q}$, the parameters $\lambda_{\mathcal{M}}$ and $\lambda_{\mathcal{A}}$ need to be selected by the user. We now study the sensitivity of the ULTRA-V performance to variations of the parameters within the parameter search intervals presented in Section IV-A. Fig. 3 shows the values of $\text{MSE}_{\mathcal{A}}$ resulting from unmixing the data using each combination of the parameter values. The sensitivity clearly tends to increase when values less than 1 are used for both parameters. Our practical experience indicates that good $\text{MSE}_{\mathcal{A}}$ results can be obtained using $\lambda_{\mathcal{M}}$ in $[0, 1]$ and large values about 100 for $\lambda_{\mathcal{A}}$. Moreover, some insensitivity is verified for small changes in $\lambda_{\mathcal{A}}$ about large values. Thus, searching $\lambda_{\mathcal{A}}$ in $[0.001, 100]$ with values spaced by decades as done for the examples in Section IV-A seems reasonable.

2) Discussion: A close look at Fig. 2 reveals the abundance maps estimated using ULTRA-V look noisier than those obtained using ELMM and GLMM. This is because the proposed approach does not impose the local smoothness imposed by total variation (TV) but emphasizes abundance regularity by enforcing a low-rank property. We note, however, that the spacial smoothness imposed by TV is not necessarily mandatory for a good abundance estimation, as can be verified from the results in Table I.

B. Real Data

For the simulations with real data, we considered three data sets, consisting of the Houston, Samson, and Jasper Ridge images. All data sets were captured by the AVIRIS, which originally has 224 spectral bands. For all images, the water absorption bands were removed resulting in 188 bands for the Houston image, 156 bands for the Samson image, and 198 bands for the Jasper Ridge image. The Houston data set is known to have four predominant EMs [11], [43]. The Samson and Jasper Ridge images are known to have three and four EMs, respectively [44]. For all images, the EMs were extracted using the VCA [40]. Fig. 4 shows the reconstructed abundance maps for all images and for all tested methods. The quantitative results are given in Table II. Note that since the ground truth (correct) abundance values are not available for

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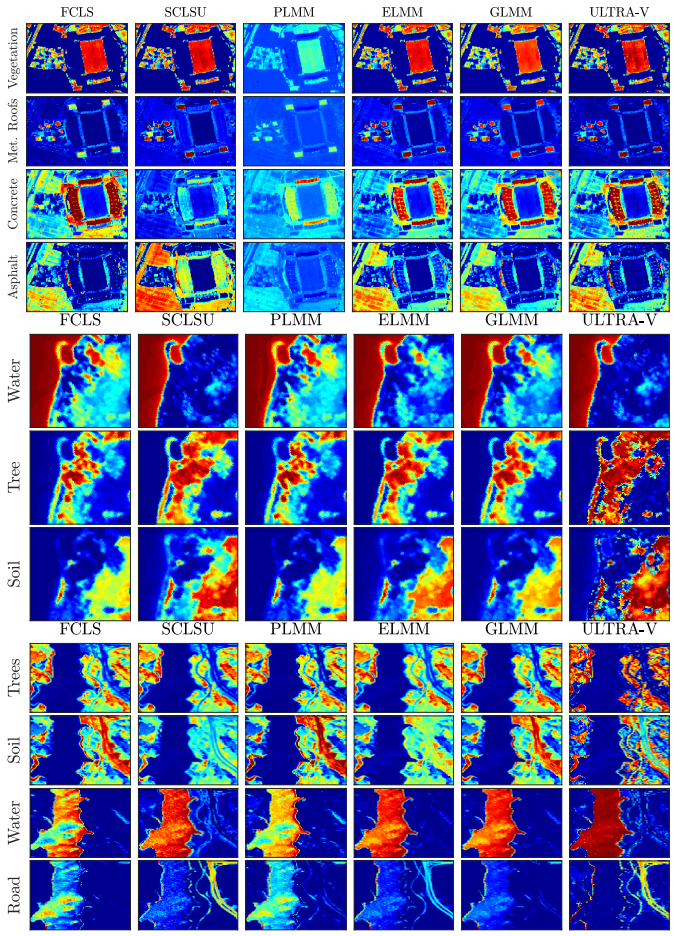


Fig. 4. (Top) Abundance maps of the Houston, (middle) Samson, and (bottom) Jasper Ridge data sets for all tested algorithms. Abundance values represented by colors ranging from blue ($\alpha_k = 0$) to red ($\alpha_k = 1$).

these images, only the reconstruction error $\text{MSE}_{\mathcal{R}}$ has been used as a sort of quality verification.

The last column in Fig. 4 shows that the proposed ULTRA-V method provided an accurate abundance estimation, clearly outperforming the competing algorithms.[1] In fact, for the Concrete and Metallic Roofs EMs, the ULTRA-V abundance map presents stronger Concrete and Metallic Roofs components in the stadium stands and stadium towers, respectively, when compared with the other methods equipped for dealing with spectral variability. The performance improvement provided by ULTRA-V is clearer for the Samson and Jasper Ridge images. For instance, there is significantly less confusion between the Water, Tree, and Soil EMs in the ULTRA-V results for the Samson image when compared to those of the PLMM, ELMM, and GLMM methods. Similarly, the ULTRA-V reconstructed abundance maps of the Jasper Ridge image show a much stronger Water component in the river and less confusion between the Tree, Soil, and Road EMs.

The objective metrics presented in Table II indicate that ULTRA-V yields competitive reconstruction errors in terms of MSE. These results, however, should be interpreted with proper care, as the connection of reconstruction error and abundance estimation is not straightforward.

[1]The differences between the ELMM and ULTRA-V results are less significant for the Houston image.

TABLE II
REAL DATA

Algorithm	Houston Data		Samson		Jasper Ridge	
	$\text{MSE}_{\mathcal{R}}$	Time	$\text{MSE}_{\mathcal{R}}$	Time	$\text{MSE}_{\mathcal{R}}$	Time
FCLS	0.2283	1.90	0.0177	1.38	0.3567	1.59
SCLS	0.0037	2.04	0.0041	1.29	0.0271	1.79
PLMM	0.0190	454.90	0.0034	105.36	0.0257	72.86
ELMM	**0.0010**	474.45	7.82e-4	40.50	0.0058	100.49
GLMM	**1.0e-5**	1326.85	**0.2e-5**	50.62	**2.5e-5**	214.07
ULTRA-V	0.0018	264.71	**6.4e-5**	148.46	**15.0e-5**	120.91

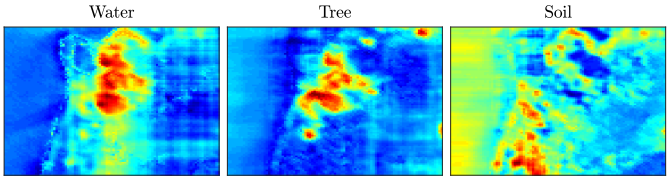


Fig. 5. Average of the ULTRA-V EMs tensor projection over the three principal components for the Samson data set.

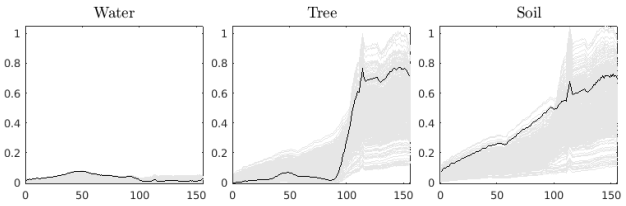


Fig. 6. VCA result (black) and ULTRA-V (gray) EMs for each pixel of the Samson data set.

The execution times in Table II indicate that, as discussed in Section III-E, ULTRA-V did not scale well with the larger image sizes and higher number of EMs, which directly impacted the CPD stage of ULTRA-V. Moreover, the more complex images resulted in higher rank estimates using the strategy discussed in Section III-F. This indicates that there is still room for improving the proposed method by either providing a segmentation strategy or using faster CPD methods. This, however, is an open problem that will be addressed in future works.

To assess the estimated EM variabilities, we analyzed the results for the Samson data set. We considered two approaches. The first approach consisted in averaging the projection of the estimated EMs on the three eigenvectors associated with the three largest eigenvalues for each EM. The results are shown in Fig. 5. These plots illustrate the EM variances for each pixel, with red implying a large variance and blue a small variance. The second approach consisted in directly comparing the EMs estimated with ULTRA-V and VCA. The results are shown in Fig. 6. These figures illustrate the ability of the proposed method to characterize the spectral variability while enforcing a spatial structure for the estimated EMs.

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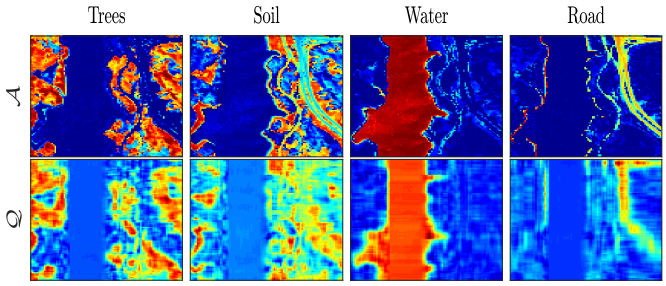


Fig. 7. Comparison of tensors $\mathcal{A}$ and $\mathcal{Q}$ after ULTRA-V convergence for the Jasper Ridge data set.

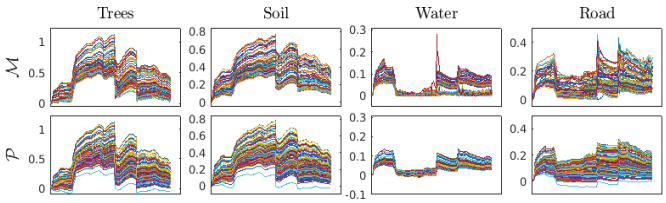


Fig. 8. Comparison of tensors $\mathcal{M}$ and $\mathcal{P}$ after ULTRA-V convergence for the Jasper Ridge data set.

To illustrate the role of the low-rank tensors $\mathcal{P}$ and $\mathcal{Q}$, we compare them to the abundances $\mathcal{A}$ and EMs $\mathcal{M}$ in Figs. 7 and 8, for the Jasper Ridge data set. Fig. 7 shows the estimated abundances $\mathcal{A}$ and their low-rank counterpart $\mathcal{Q}$. One can verify that $\mathcal{Q}$ (bottom row) has a very coarse spatial distribution when compared with $\mathcal{A}$ (top row). This shows that imposing the low-rank structure through a regularization constraint gives the resulting abundances enough flexibility to model fine-scale spatial details while maintaining most of its spatial distribution. Fig. 8 leads to similar conclusions for the EMs. One can note that the low-rank tensor $\mathcal{P}$ has a coarser structure when compared with the estimated EMs $\mathcal{M}$. This distinction can be seen very clearly for the Water and Road EMs.

V. Conclusion

In this article, we proposed a new low-rank regularization strategy for introducing low-dimensional spatial–spectral structure into the abundance and EM tensors for HU considering spectral variability. The resulting iterative algorithm, called ULTRA-V, imposes a low-rank structure by means of regularizations that force most of the energy of the estimated abundances and EMs to lay within a low-dimensional structure. The proposed approach does not confine the estimated abundances and EMs to a strict low-rank structure, which would not adequately account for the complexity experienced in real-world scenarios. The proposed methodology includes also a strategy to estimate the rank of the regularization tensors $\mathcal{P}$ and $\mathcal{Q}$, leaving only two parameters to be adjusted within a relatively reduced search space. Simulation results using both synthetic and real data showed that the ULTRA-V can outperform state-of-the-art unmixing algorithms accounting for spectral variability.

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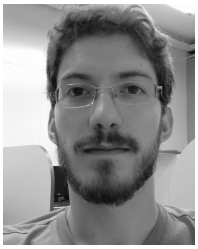
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